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(54) **SYSTEMS AND METHODS FOR HIGH-VOLTAGE-TOLERANT LEVEL SHIFTERS**

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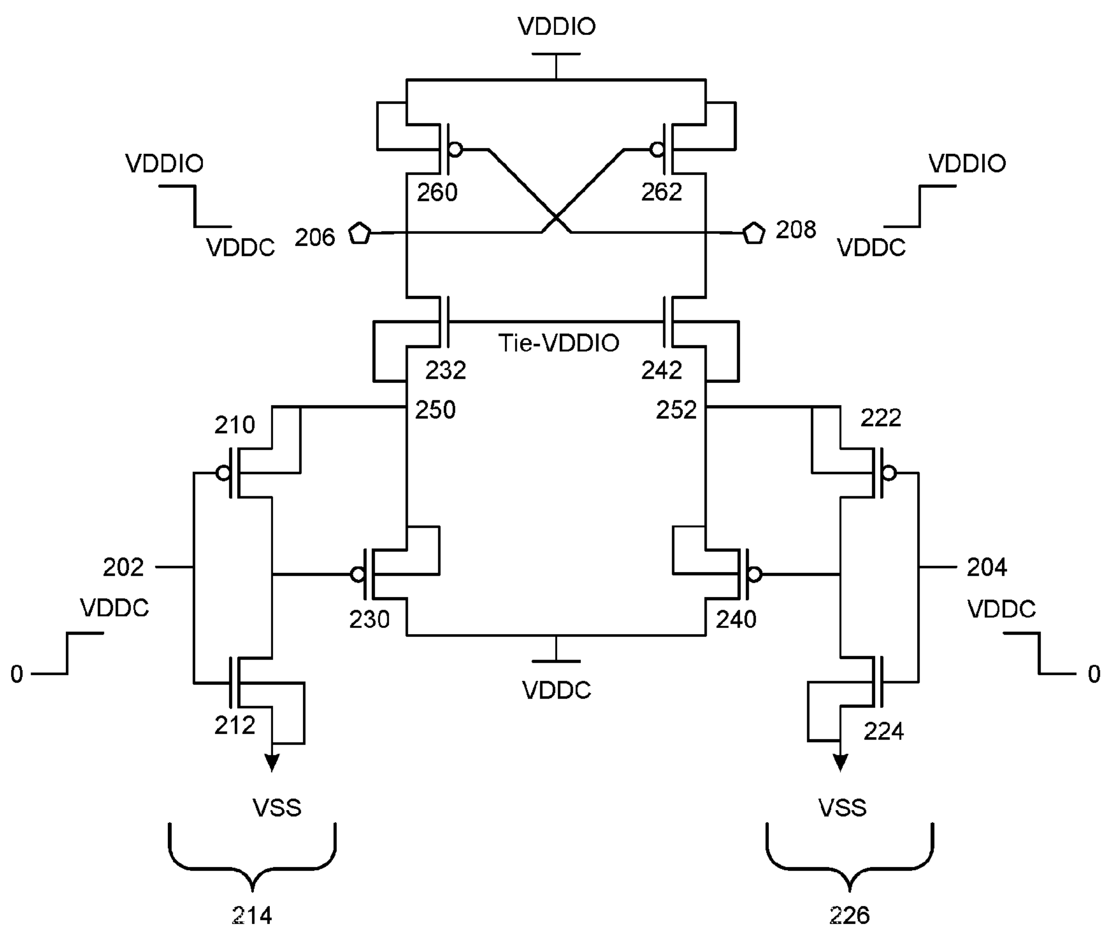
CPC *H03K 19/018521* (2013.01)

(57) **ABSTRACT**

The disclosed device for high-voltage-tolerant level shifters can include a level shifter device configured to shift an input signal from an input voltage range to an output voltage range, where a span from a bottom of the input voltage range to a top of the output voltage range is greater than a voltage differential tolerance of any of the plurality of transistors. Various other devices and systems are also disclosed.

200



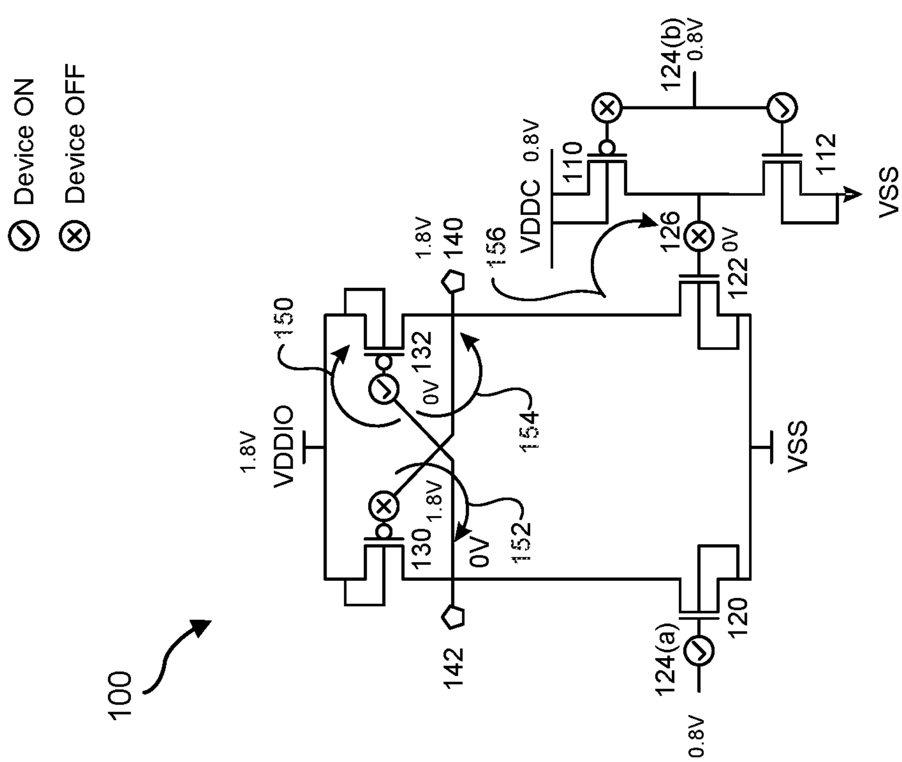

Device ON
Device OFF
100
VDDIO
1.8V
150
130
132
0V
1.8V
152
154
142
140
1.8V
124(a)
0.8V
120
122
156
126
0V
VDDC
0.8V
110
112
124(b)
0.8V
VSS
VSS

FIG. 1A

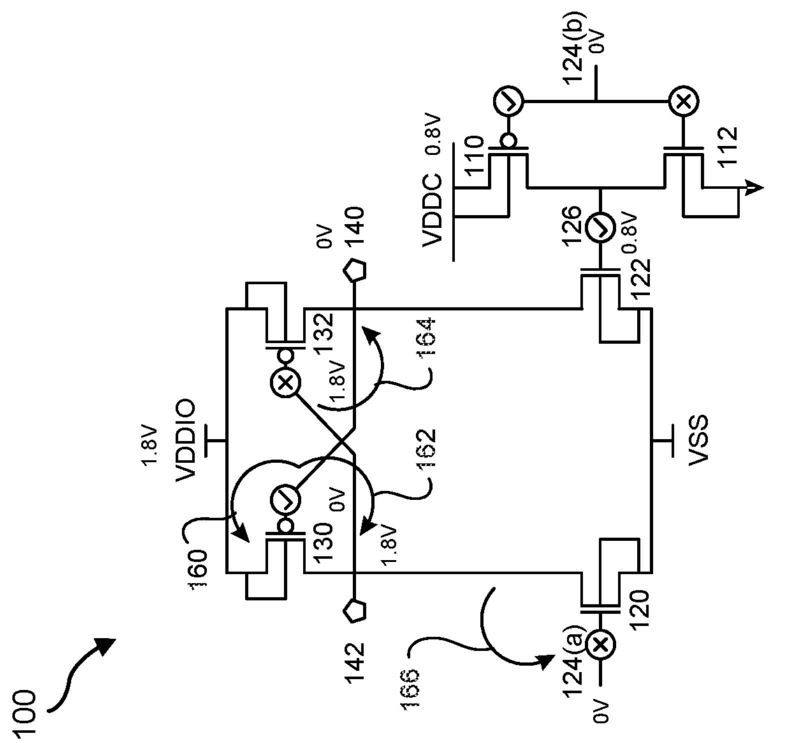

100
VDDIO
1.8V
160
162
164
130
132
0V
1.8V
1.8V
142
140
0V
166
124(a)
0V
120
122
126
0.8V
VDDC
0.8V
110
112
124(b)
0V
VSS

FIG. 1B

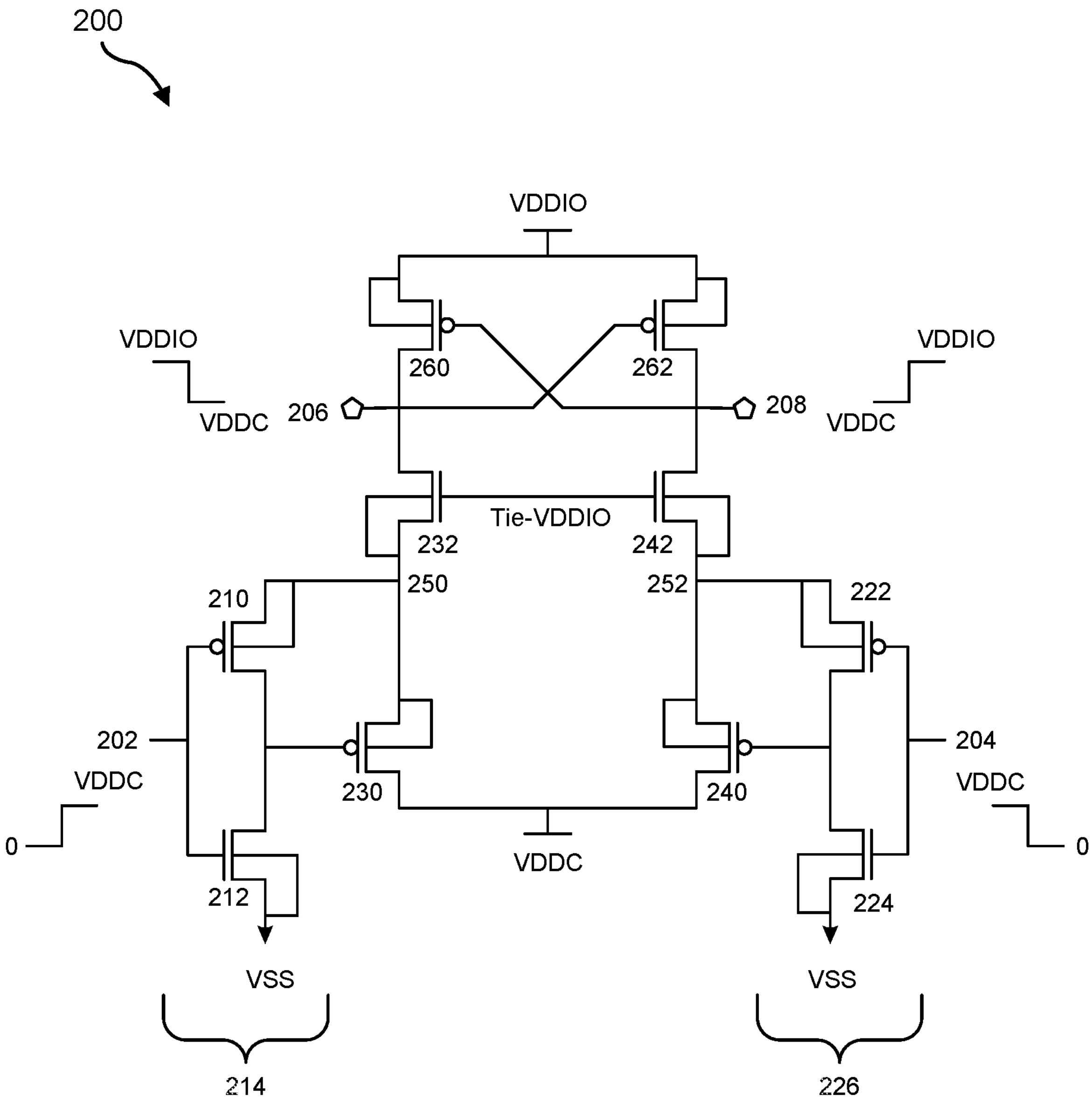

200
VDDIO
VDDIO
VDDC
206
260
262
208
VDDIO
VDDC
Tie-VDDIO
232
242
250
252
210
222
202
204
VDDC
230
240
VDDC
0
0
VDDC
212
224
VSS
VSS
214
226

FIG. 2

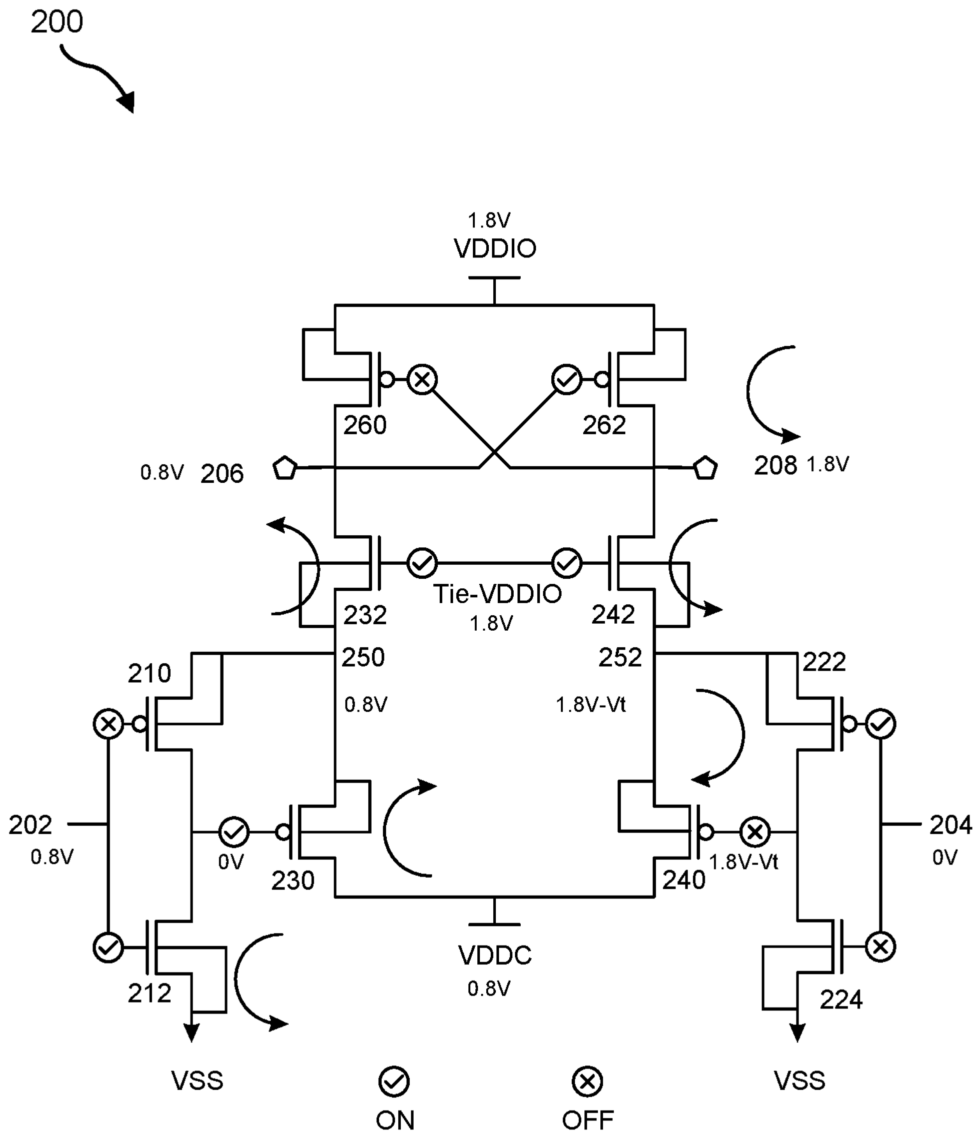

200
1.8V
VDDIO
260
262
0.8V 206
208 1.8V
232
Tie-VDDIO
1.8V
242
250
252
210
222
0.8V
1.8V-Vt
202
0.8V
0V
230
240
1.8V-Vt
204
0V
212
224
VDDC
0.8V
VSS
ON
OFF
VSS

FIG. 3

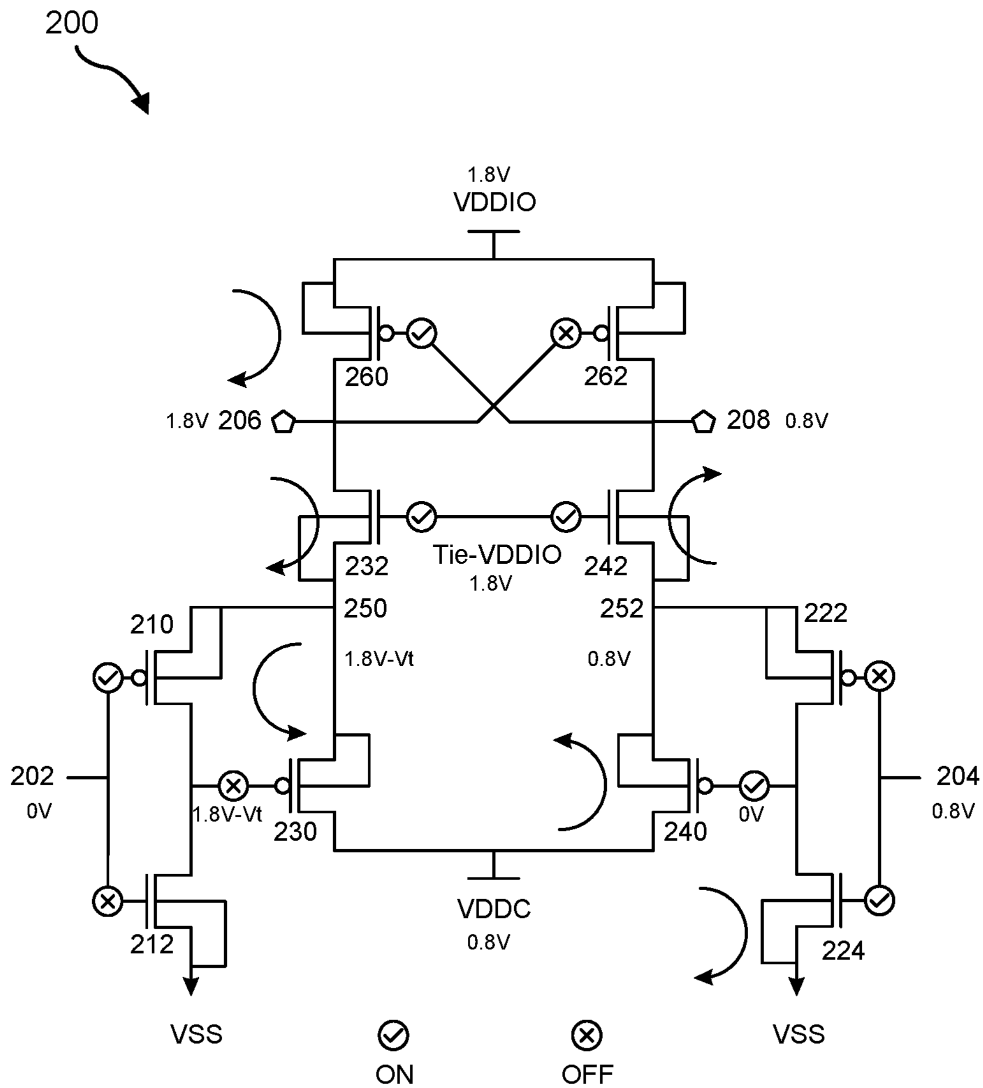

200
1.8V
VDDIO
260
262
1.8V 206
208 0.8V
232
Tie-VDDIO
1.8V
242
250
252
210
222
1.8V-Vt
0.8V
202
0V
1.8V-Vt
230
240
0V
204
0.8V
212
VDDC
0.8V
224
VSS
ON
OFF
VSS

FIG. 4

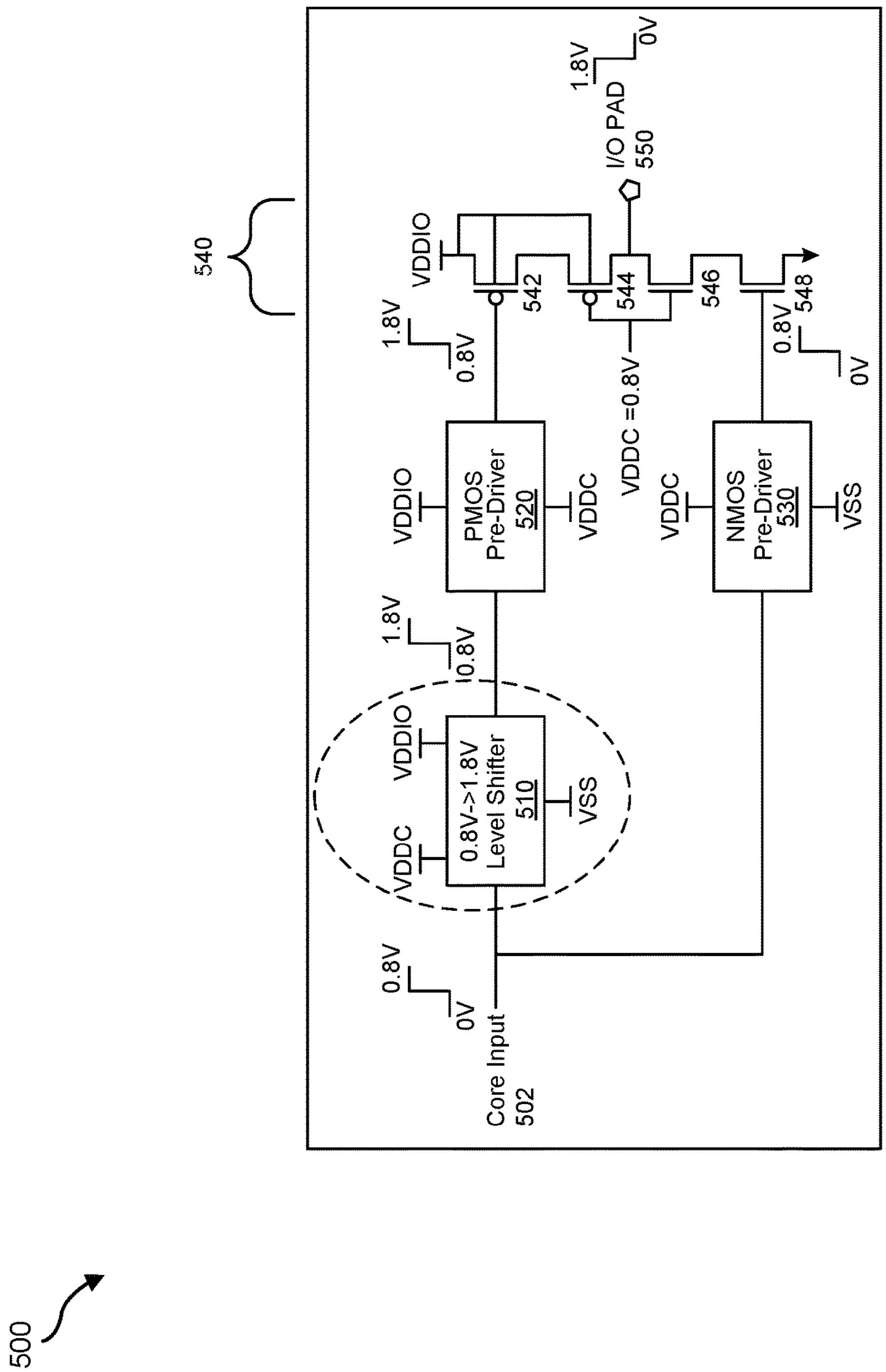

500
540
Core Input
502
0.8V
0V
0.8V->1.8V
Level Shifter
510
VDDC
VDDIO
VSS
1.8V
0.8V
PMOS
Pre-Driver
520
VDDIO
VDDC
NMOS
Pre-Driver
530
VDDC
VSS
1.8V
0.8V
VDDIO
542
544
546
548
VDDC =0.8V
0.8V
0V
I/O PAD
550
1.8V
0V

FIG. 5

SYSTEMS AND METHODS FOR HIGH-VOLTAGE-TOLERANT LEVEL SHIFTERS

BRIEF DESCRIPTION OF THE DRAWINGS

[0001] The accompanying drawings illustrate a number of example embodiments and are a part of the specification. Together with the following description, these drawings demonstrate and explain various principles of the present disclosure.

[0002] FIG. **1**A is a diagram of an example level shifter in a first state.

[0003] FIG. **1**B is a diagram of the example level shifter of FIG. **1**A in a second state.

[0004] FIG. **2** is a diagram of an example high-voltage-tolerant level shifter.

[0005] FIG. **3** is a diagram of the example high-voltage-tolerant level shifter of FIG. **2** in a first state.

[0006] FIG. **4** is a diagram of the example high-voltage-tolerant level shifter of FIG. **2** in a second state.

[0007] FIG. **5** is a diagram of an example system including a high-voltage tolerant level shifter.

[0008] Throughout the drawings, identical reference characters and descriptions indicate similar, but not necessarily identical, elements. While the example embodiments described herein are susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and will be described in detail herein. However, the example embodiments described herein are not intended to be limited to the particular forms disclosed. Rather, the present disclosure covers all modifications, equivalents, and alternatives falling within the scope of the appended claims.

DETAILED DESCRIPTION OF EXAMPLE IMPLEMENTATIONS

[0009] The present disclosure is generally directed to devices and systems for high-voltage-tolerant voltage level shifters. Advancements in computing hardware continue to put pressure on increasing processing power, increasing power efficiency, increasing component density, reducing cost, and increasing reliability. Reducing the minimum feature size of transistors can advance these objectives. However, devices conforming a smaller feature size may have other limitations. For example, a device with a smaller feature size may have a lower tolerance of voltage differential across the device. Thus, for example, I/O devices with a 6 nanometer (nm) technology node minimum feature size may be 1.8 volt (V) tolerant, while the same I/O devices with a 5 nm technology node minimum feature size may only be 1.5 V tolerant. An attempt to use 1.5 V tolerant I/O devices in a high-voltage application (e.g., relating to the I/O voltage domain signal, which may span from 0 V to 1.8 V) may result in over-voltage stress on the 1.5 V tolerant devices.

[0010] A voltage level converter circuit (or "level shifter") may convert a signal from a source voltage domain to a destination voltage domain in a multi-voltage domain system. Devices and systems described herein may include high-voltage tolerant level shifters that help to eliminate over-voltage stress issues that would otherwise be experienced by low-voltage-tolerant transistors when operating in high-voltage applications.

[0011] For example, one or more of the devices described herein may be implemented as a level shifter composed of (and, e.g., interoperating with) 5 nm technology node and I/O devices that converts a Core voltage domain signal (e.g., about 0 V to 0.8 V) to an I/O voltage domain signal (e.g., about 0 V to 1.8 V) where the voltage tolerance of the 5 nm technology node and I/O devices is less than 1.8 V (e.g., 1.5 V). However, as will be explained in greater detail below, the level shifter described herein may successfully convert a signal from the Core domain to the I/O domain without over-voltage stress to the 5 nm technology node and I/O devices.

[0012] In addition, in some examples, the level shifter devices described herein may be relatively simple, space efficient, high performance, and/or power efficient. For example, one or more of the level shifter devices described herein may optionally avoid and/or exclude elements that may consume a significant amount of area, such as operational amplifiers, resistors, and diodes. Furthermore, these level shifter devices may optionally avoid and/or exclude the use of techniques that may demand more static power and, thus, be less power efficient, such as intermediate voltage generation or biasing. Furthermore, these level shifter devices may optionally avoid and/or exclude approaches that may degrade the performance of the circuit, such as device-stacking techniques that limit the voltage across any two terminals of a device and may thereby degrading performance during high-speed switching.

[0013] Furthermore, one or more of the level shifter devices described herein may mitigate device reliability issues, such as gate-oxide breakdown, which may be caused due to the application of a high voltage at gate terminals of devices beyond their tolerance limits. Additionally, in some examples, one or more of these level shifter devices may avoid leakage current and/or latch-up, by, for example, bulk connections that avoid forward-biased parasitic diodes in the transistors. Moreover, one or more of these level shifter devices may minimize contention by avoiding any static current paths in the circuit while switching. In addition, one or more of these level shifter devices may provide a balanced rise/fall slew at the output, thereby making them suitable to work in both the data path and the clock path at high frequencies (e.g., several hundreds of megahertz (MHz)). Furthermore, these level shifter devices may be useful for a wide range of Core voltage domains (e.g., including, without limitation, Core domains with maximums ranging anywhere from 0.6 V to 1.25 V) and I/O voltage domains (e.g., including, without limitation, I/O domains with maximums ranging anywhere from 1.62 V to 1.98 V).

[0014] FIGS. **1**A and **1**B are diagrams of an example level shifter **100** in two states of operation. By way of example, level shifter **100** may be placed in a context to shift from a Core domain (0-0.8 V) to an I/O domain (0-1.8 V). In addition, the transistors of level shifter **100** may be implemented in 5 nm technology node, and may have a voltage tolerance of 1.5 V. Thus, FIGS. **1**A and **1**B illustrate ordinary consequences to an example level shifter implemented in 5 nm technology node.

[0015] As shown in FIGS. **1**A and **1**B, level shifter **100** may include various transistors. In some examples, transistors illustrated and/or described herein may be field-effect transistors (FETs). For example, transistors illustrated and/or described herein may be metal-oxide-semiconductor FETs (MOSFETs). In some examples, a MOSFET may be a p-type MOSFET (PMOS) or an n-type MOSFET (NMOS). For example, level shifter **100** may include PMOS devices **110**, **130**, and **132** and NMOS devices **112**, **120** and **122**. Level shifter **100** may also include inputs **124**(*a*) and **124**(*b*) and outputs **140** and **142**. Devices **110** and **112** may act as input inverters, resulting in an inverted input **126**. The pair of PMOS devices **130** and **132** may be cross-coupled.

[0016] In the first case, shown in FIG. **1**A, an input of 0.8 V may be provided to level shifter **100** (at inputs **124**(*a*) and **124**(*b*)). The node above input **124**(*a*) may be pulled down to 0 V. Thus, device **132** may experience a voltage differential **150** of |0 V-1.8 V|=1.8 V, and another voltage differential **154** of |0 V-1.8 V|=1.8 V, each greater than the voltage differential tolerance of 1.5 V. Similarly, where input **124**(*b*) is 0.8 V, inverted input **126** may be 0 V. Thus, device **122** may experience a voltage differential **156** of |0 V-1.8 V|=1.8 V, greater than the voltage differential tolerance of 1.5 V. In addition, device **130** may experience a voltage differential **152** of |1.8 V-0 V|=1.8 V, greater than the voltage differential tolerance of 1.5 V.

[0017] In the second case, shown in FIG. **1**B, an input of 0 V may be provided to level shifter **100** (at inputs **124**(*a*) and **124**(*b*)). Thus, device **130** may experience a voltage differential **160** of |0 V-1.8 V|=1.8 V, and another voltage differential **162** of |0 V-1.8 V|=1.8 V, each greater than the voltage differential tolerance of 1.5 V. Similarly, where input **124**(*b*) is 0 V, inverted input **126** may be 0.8 V. Thus, device **132** may experience a voltage differential **164** of |1.8 V-0 V|=1.8 V, greater than the voltage differential tolerance of 1.5 V. In addition, device **120** may experience a voltage differential **166** of |0 V-1.8 V|=1.8 V, greater than the voltage differential tolerance of 1.5 V.

[0018] Thus, as can be appreciated from the foregoing, devices of level shifter **100** may experience electrical overstress during typical operation.

[0019] FIG. **2** is a diagram of an example high-voltage-tolerant level shifter **200**. As shown in FIG. **2**, level shifter **200** may include various transistors. For example, level shifter **200** may include PMOS devices **210**, **222**, **230**, **240**, **260**, and **262**. In addition, level shifter **200** may include NMOS devices **212**, **224**, **232**, and **242**. Level shifter **200** may also include an input **202**, an inverted input **204**, an output **208**, and an inverted output **206**. Devices **210** and **212** may form a part of an input buffer stage **214** and devices **222** and **224** may form a part of an input buffer stage **226**. Devices **260** and **262** may be cross-connected, and devices **230** and **240** may act as a driving pair of devices that drive nets **250** and **252**. In some examples, devices **232** and **242** may have a high threshold voltage (Vt). For example, devices **232** and **242** may have a high channel length, resulting in a higher threshold voltage. Additionally or alternatively, devices **232** and **242** may represent stacked devices, resulting in a higher threshold voltage.

[0020] As shown in FIG. **2**, the input voltage range of level shifter **200** may correspond to the Core voltage domain—e.g., from 0 V (ground) to VDDC, and the output voltage range of level shifter **200** may be from VDDC to the I/O voltage domain (VDDIO). Thus, level shifter **200** may handle both the input voltage range of 0 V to VDDC and the output voltage range of VDDC to VDDIO. Taking, by way of example, VDDC to be 0.8 V and VDDIO to be 1.8 V, the total span of the voltage ranges handled by level shifter **200** may be 1.8 V. Nevertheless, in some examples, each of the devices of level shifter **200** may be individually tolerant to only a voltage difference less than 1.8V. For example, the devices of level shifter **200** may be of a 5 nm technology node or lower design, and may only be tolerant to a voltage differential of 1.5 V.

[0021] As can be seen in FIG. **2**, and in contrast to the design of level shifter **100** of FIGS. **1**A and **1**B, the source terminals of devices **230** and **240** may be connected to VDDC (the input voltage domain). By way of example, the maximum voltage level of the Core voltage domain may be 0.8 V. Thus, as will be explained in greater detail below, as devices **230** and **240** drive nets **250** and **252**, no device in level shifter **200** will experience a greater voltage differential greater than 1.5 V.

[0022] In some examples, and in contrast to the design of level shifter **100**, the source and the bulk of device **210** may connect to a drain net **250** (which is connected to the drain of device **230**). Likewise, the source and the bulk of device **222** may connect to a drain net **252** (which is connected to the drain of device **240**). This configuration may help to keep the parasitic bulk diode of devices **210** and **222** in a reverse bias condition and avoid any undesired flow of current in the substrate. In addition, this configuration may avoid power sequencing scenarios where VDDC is present and the Input/Output voltage source (VDDIO) is absent, or where VDDIO is present and VDDC is absent. By not connecting the source and bulk of devices **210** and **222** directly to VDDC, level shifter **200** may avoid stress situations in power sequencing scenarios. Thus, level shifter **200** may be power sequencing independent.

[0023] FIG. **3** is a diagram of high-voltage-tolerant level shifter **200** of FIG. **2** in a first scenario. As shown in FIG. **3**, an input at **202** may be 0.8 V and an inverted input at **204** may be 0 V. Because device **212** is activated, device **230** may be enabled. Device **230** may assign VDDC (0.8V) to net **250**, and device **232** may transfer 0.8 V to inverted output **206** (as the gate terminal for device **232** is at Tie-VDDIO, which may be 1.8 V). As the signal at output **206** is 0.8 V and it is connected to the gate of device **262**, device **262** may be enabled and activated as its source terminal is at 1.8 V and may assign 1.8 V to output **208**. Device **260** may be turned off as its gate and source terminals are at the same potential. Because both the gate and source terminals of device **242** are at 1.8 V, device **242** may transfer 1.8 V-Vt to net 252 (where Vt is the threshold voltage of the device). As may be appreciated, Vt may be high enough such that the difference between 1.8 V and Vt is less than or equal to the tolerance limit of the device (e.g., less than or equal to 1.5 V). Thus, in this example, Vt may be greater than or equal to 0.3 V. Since the signal at inverted input **204** is 0V, device **222** may be turned on and send 1.8 V-Vt to the gate terminal of device **240**, thereby turning it off. As can be appreciated, none of the devices experience a voltage differential of greater than 1.5 V, even though various parts of level shifter **200** may be at 0 V or 1.8 V.

[0024] FIG. **4** is a diagram of the high-voltage-tolerant level shifter **200** of FIG. **2** in a second scenario. As shown in FIG. **4**, a signal at inverted input **204** may be 0.8 V and a signal at input **202** may be 0 V. Because device **224** is activated, device **240** may be enabled. Device **240** may assign VDDC (0.8 V) to net **252**, and device **242** may transfer 0.8 V to output **208** (as the gate terminal for device **242** is at Tie-VDDIO, which may be 1.8 V). As the signal at output **208** is 0.8 V and it is connected to the gate of device **260**, device **260** may be enabled and activated as its source terminal is at 1.8 V and may assign 1.8 V to inverted output **206**. Device **262** may be turned off as its gate and source terminals are at the same potential. Because both the gate and source terminals of device **232** are at 1.8 V, device **232** may transfer 1.8 V-Vt to net **250**. As may be appreciated, Vt may be high enough such that the difference between 1.8 V and Vt is less than or equal to the tolerance limit of the device (e.g., less than or equal to 1.5 V). Thus, in this example, Vt may be greater than or equal to 0.3 V. Since the signal at input **202** is 0 V, device **210** may be turned on and send 1.8 V-Vt to the gate terminal of device **230**, thereby turning it off. As can be appreciated, none of the devices experience a voltage differential of greater than 1.5 V, even though various parts of level shifter **200** may be at 0 V or 1.8 V.

[0025] FIG. **5** is a diagram of an example system **500** including a high-voltage tolerant level shifter **510**. In some examples, high-voltage tolerant level shifter **510** may be level shifter **200** of FIG. **2**. As shown in FIG. **5**, system **500** may include complementary pre-drivers (e.g., a PMOS pre-driver **520** and an NMOS pre-driver **530**). System **500** may also include a output driver circuit **540**, which may include PMOS devices **542** and **544** and NMOS devices **546** and **548**. System **500** may receive a core input **502** and provide a level shifted signal to an I/O PAD **550**. For example, core input **502** may be provided as input to level shifter **510** and to NMOS pre-driver **530**. Level shifter **510** may shift the signal from a 0 V-0.8 V range to a 0.8 V to 1.8 V range. In some examples, the devices illustrated in system **500** of FIG. **5**, including transistors used to implement level shifter **510**, may not be tolerant of a voltage differential of 1.8 V, but level shifter **510** may shift the signal without over-stress on the devices within level shifter **510** or other devices within system **500**. Level shifter **510** may thus produce a signal in a range from 0.8 V to 1.8 V. Output driver circuit **540** may provide the signal, in a range from 0 V to 1.8 V, to I/O PAD **550** without exposing any of devices **542**, **544**, **546**, and **548** to a voltage differential of greater than 1.5 V.

[0026] A method of manufacture of a high-voltage tolerant level shifter may include dispose a plurality of transistors to form a level-shifter device (e.g., according to one or more of the layouts illustrated and/or described above) that is configured to shift an input signal from an input voltage range to an output voltage range, where a span from a bottom of the input voltage range to a top of the output voltage range is greater than a voltage differential tolerance of any of the plurality of transistors.

[0027] While the foregoing disclosure sets forth various implementations using specific block diagrams, flowcharts, and examples, each block diagram component, flowchart step, operation, and/or component described and/or illustrated herein can be implemented, individually and/or collectively, using a wide range of hardware, software, or firmware (or any combination thereof) configurations. In addition, any disclosure of components contained within other components should be considered example in nature since many other architectures can be implemented to achieve the same functionality.

[0028] The process parameters and sequence of steps described and/or illustrated herein are given by way of example only and can be varied as desired. For example, while the steps illustrated and/or described herein can be shown or discussed in a particular order, these steps do not necessarily need to be performed in the order illustrated or discussed. The various example methods described and/or illustrated herein can also omit one or more of the steps described or illustrated herein or include additional steps in addition to those disclosed.

[0029] While various implementations have been described and/or illustrated herein in the context of fully functional computing systems, one or more of these example implementations can be distributed as a program product in a variety of forms, regardless of the particular type of computer-readable media used to actually carry out the distribution. The implementations disclosed herein can also be implemented using modules that perform certain tasks. These modules can include script, batch, or other executable files that can be stored on a computer-readable storage medium or in a computing system. In some implementations, these modules can configure a computing system to perform one or more of the example implementations disclosed herein.

[0030] The preceding description has been provided to enable others skilled in the art to best utilize various aspects of the example implementations disclosed herein. This example description is not intended to be exhaustive or to be limited to any precise form disclosed. Many modifications and variations are possible without departing from the spirit and scope of the present disclosure. The implementations disclosed herein should be considered in all respects illustrative and not restrictive. Reference should be made to the appended claims and their equivalents in determining the scope of the present disclosure.

[0031] Unless otherwise noted, the terms "connected to" and "coupled to" (and their derivatives), as used in the specification and claims, are to be construed as permitting both direct and indirect (i.e., via other elements or components) connection. In addition, the terms "a" or "an," as used in the specification and claims, are to be construed as meaning "at least one of." Finally, for ease of use, the terms "including" and "having" (and their derivatives), as used in the specification and claims, are interchangeable with and have the same meaning as the word "comprising."

What is claimed is:

1. A level-shifter device comprising a plurality of transistors, wherein the level-shifter device is configured to shift an input signal from an input voltage range to an output voltage range, wherein a span from a bottom of the input voltage range to a top of the output voltage range is greater than a voltage differential tolerance of any of the plurality of transistors.

2. The level-shifter device of claim **1**, wherein the plurality of transistors comprises a pair of cross-coupled field-effect transistors, wherein a pair of drain nets of the pair of cross-coupled field-effect transistors comprise a pair of complementary output nets of the level-shifter device.

3. The level-shifter device of claim **2**, wherein a voltage source of the pair of cross-coupled field-effect transistors provides a voltage level at the top of the output voltage range.

4. The level-shifter device of claim **2**, wherein the plurality of transistors comprises a pair of driving field-effect transistors with high voltage thresholds that activate the pair of cross-coupled field-effect transistors.

5. The level-shifter device of claim **4**, wherein a voltage source of the pair of driving field-effect transistors provides a voltage level at the bottom of the output voltage range.

6. The level-shifter device of claim **4**, further comprising a pair of input buffers;

wherein the pair of input buffers each comprise a buffering field-effect transistor, wherein a source and a bulk of the buffering field-effect transistor connects to a corresponding drain net of a corresponding transistor of the pair of driving field-effect transistors.

7. The level-shifter device of claim **1**, wherein:

the input voltage range is from a ground voltage level to a core voltage level; and

the output voltage range is from a core voltage level to an input/output voltage level.

8. The level-shifter device of claim **7**, wherein:

the core voltage level is about 0.8 volts; and

the input/output voltage level is about 1.8 volts.

9. The level-shifter device of claim **7**, wherein the voltage differential tolerance is about 1.5 volts.

10. The level-shifter device of claim **1**, wherein none of the plurality of transistors experiences voltage stress greater than the voltage differential tolerance when the level-shifter device is in operation.

11. A system comprising:

a level-shifter device comprising a plurality of transistors, wherein the level-shifter device is configured to shift an input signal from an input voltage range to an output voltage range, wherein a span from a bottom of the input voltage range to a top of the output voltage range is greater than a voltage differential tolerance of any of the plurality of transistors;

a first pre-driver and a second pre-driver that is complementary to the first pre-driver, wherein an output of the level-shifter device is an input to the first pre-driver and the input signal is an input to the second pre-driver; and

an output buffer circuit that receives outputs of the first and second pre-drivers as inputs and outputs a conditioned signal to an input/output pad.

12. The system of claim **11**, wherein the plurality of transistors comprises a pair of cross-coupled field-effect transistors, wherein a pair of drain nets of the pair of cross-coupled field-effect transistors comprise a pair of complementary output nets of the level-shifter device.

13. The system of claim **12**, wherein a voltage source of the pair of cross-coupled field-effect transistors provides a voltage level at the top of the output voltage range.

14. The system of claim **12**, wherein the plurality of transistors comprises a pair of driving field-effect transistors that activate the pair of cross-coupled field-effect transistors.

15. The system of claim **14**, wherein a voltage source of the pair of driving field-effect transistors provides a voltage level at the bottom of the output voltage range.

16. The system of claim **14**, further comprising a pair of input buffer stages;

wherein the pair of input buffer stages each comprise a buffering field-effect transistor, wherein a source and a bulk of the buffering field-effect transistor connects to a corresponding drain net of a corresponding transistor of the pair of driving field-effect transistors.

17. The system of claim **11**, wherein:

the input voltage range is from a ground voltage level to a core voltage level; and

the output voltage range is from a core voltage level to an input/output voltage level.

18. The system of claim **17**, wherein:

the core voltage level is about 0.8 volts; and

the input/output voltage level is about 1.8 volts.

19. The system of claim **17**, wherein the voltage differential tolerance is about 1.5 volts.

20. A method of manufacture, comprising disposing a plurality of transistors to form a level-shifter device that is configured to shift an input signal from an input voltage range to an output voltage range, wherein a span from a bottom of the input voltage range to a top of the output voltage range is greater than a voltage differential tolerance of any of the plurality of transistors.

* * * * *